Harsha Matta

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Education

University of North Carolina at Chapel Hill — Chapel Hill, NC
B.S. in Applied Mathematics & Computer Science — *Expected May 2028*

- **Coursework:** Computer Organization, Models of Languages and Computation, Data Structures, Systems Fundamentals, Foundations of Programming, Discrete Mathematics, Linear Algebra

Technical Projects

Next: To-Do List and Calendar with AI Scheduling | *Swift, SwiftUI, AppKit, App Intents* — 2026

- Shipped to the App Store for iPhone, iPad, and Mac (universal purchase): a to-do list and calendar in one, with local-first storage in plain Markdown files, iCloud sync, and Spotlight and App Intents integration.
- Built an AI scheduling flow that proposes task placements around existing calendar commitments and writes nothing until the user approves; supports OpenAI, Anthropic, and Gemini with the user's own key, or on-device Apple Intelligence.

InOffice Tracker | *Swift, SwiftUI, SwiftData, CoreLocation, Keychain* — May – Jun 2026

- Launched on the App Store: an attendance tracker with automatic geofenced check-ins, Keychain-secured coordinates, SwiftData persistence, and CSV import/export.

Paatam | *Dart, Flutter* — Feb – Jun 2024

- Designed and shipped an app teaching the scripts of Indian languages; 275+ downloads on iOS and Android.

Quantitative Projects & Technical Writing

Volatility Risk Premium (VRP) Study | *Python, yfinance, Pandas, NumPy, Streamlit* — Ongoing

- Built a data pipeline over SPX and VIX daily closes (2010–present) that computes log returns, realized volatility, and the implied volatility premium (VIX − RV).
- Built a Streamlit dashboard with regime filtering, configurable strategy parameters (entry frequency, vol scaling, leverage cap, circuit breaker), monthly PnL heatmaps, rolling Sharpe, and regime attribution. Write-up: *Volatility Risk Premium: How to Harvest the Risk Premium?* (Jun 2026).

Options Pricing Tools & Writing | *Python, Streamlit, Pandas* — Nov 2025 – Feb 2026

- Built a Black–Scholes tool that prices historical call options and simulates cumulative payoffs, and an interactive Greeks visualizer (Delta, Gamma, Theta, Vega, Rho) across strikes and maturities.
- Wrote a three-part Medium series on option pricing, including American options via portfolio replication and the risk-neutral approach; 3,400+ total impressions.

Experience

deal.II (Open Source) — Feb 2026 – Present
Contributor, C++ Scientific Computing Library — *Remote*

- Two merged PRs to `dealii/dealii` (1.7k+ GitHub stars). Fixed a hot-path performance bug in `InterpolatedUniformGridData::value()`, used for large data tables in parallel MPI finite-element simulations, by caching the grid-spacing array instead of recomputing it on every call.
- Fixed the affine approximation in `real_to_unit_cell` for non-hypercube cells. Now working with a core maintainer on extending `GridTools::affine_cell_approximation` to simplex and mixed cells.

CyberWIN — Apr 2024 – May 2025
Intern & Marketer — *Monroe, NC*

- Led a cross-functional team building an AI document analysis feature for CMMC/NIST compliance on the CySAT platform.

Leadership

UNC Special Task Force on AI & Academic Affairs — Aug 2026 – Present
Task Force Member, Technical & Policy Track — *Chapel Hill, NC*

- Evaluate AI tools for academic research; advise senior administration on student usage guidelines and policy.

UNC Sangam — Fall 2025 – Present
Treasurer — *Chapel Hill, NC*

- Manage budgeting and expense tracking for one of UNC's largest South Asian cultural organizations, allocating funds across events including Garba and Mock Shaadi.

Skills & Honors

Languages & Frameworks: Swift, SwiftUI, AppKit, Combine, App Intents, Python, C++, Java, Dart, JavaScript
Tools & Certifications: Git, Figma, Agile/SDLC; Financial Markets (Yale/Coursera); Intro to MCP (Anthropic)
Honors & Languages: DECA State CDC Qualifier (2024); UKMT Mathematics Challenge, 2× Gold; English, Telugu (native)